INCH-POUND

MIL-DTL-16377/13B(SH)
2 August 1996
SUPERSEDING
MIL-F-16377/13A(SH)
18 March 1982

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, GENERAL LIGHTING, 180 WATTS, 120 VOLTS, 60 HERTZ, DRIPPROOF, SYMBOLS 74 AND 74.1

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-16377.

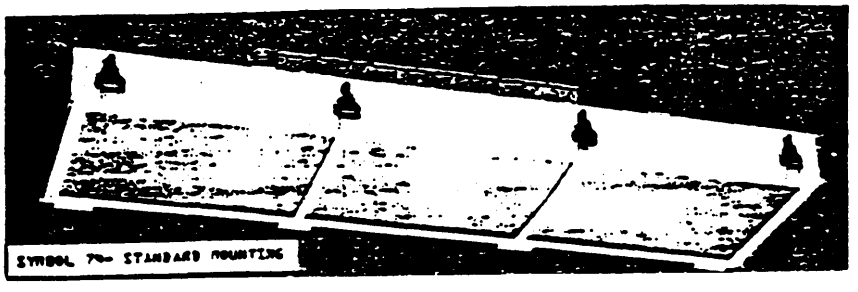


FIGURE 1. Typical view of Symbol 74.

All dimensions in inches.
For tolerances, other than specified herein, see 3.5.2 of MIL-DTL-16377.

AMSC N/A FSC 6210

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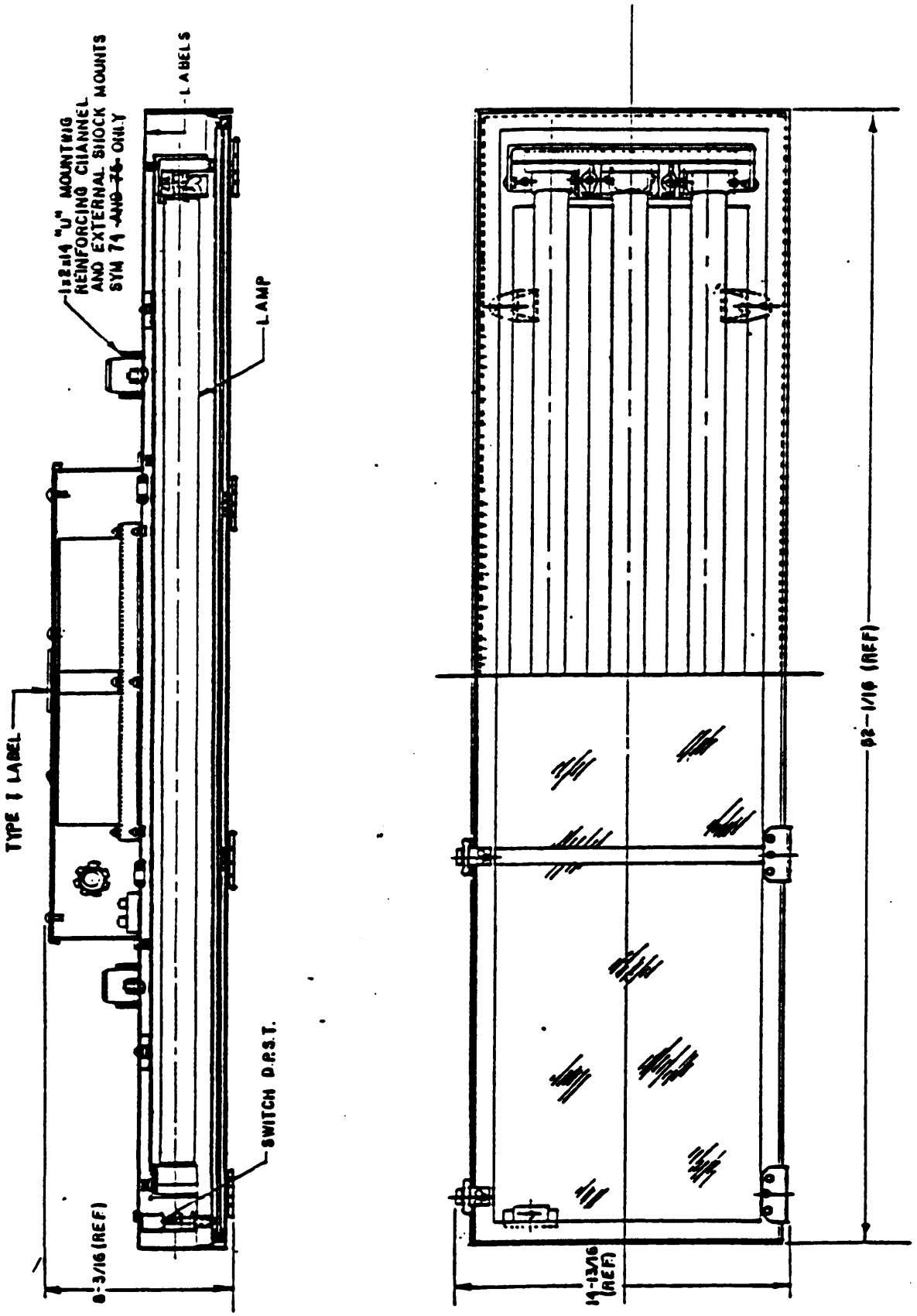


FIGURE 2. Typical views for Symbol 74.

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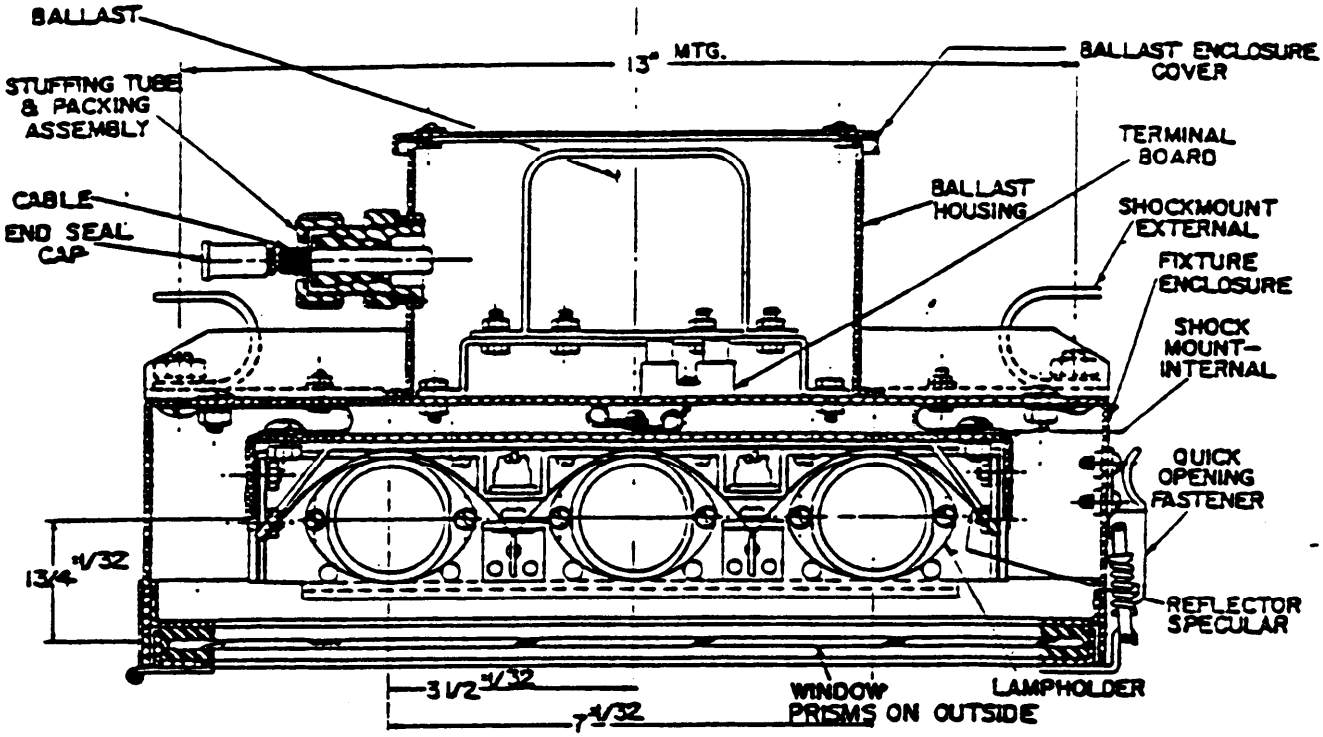


FIGURE 3. Typical section for Symbol 74.

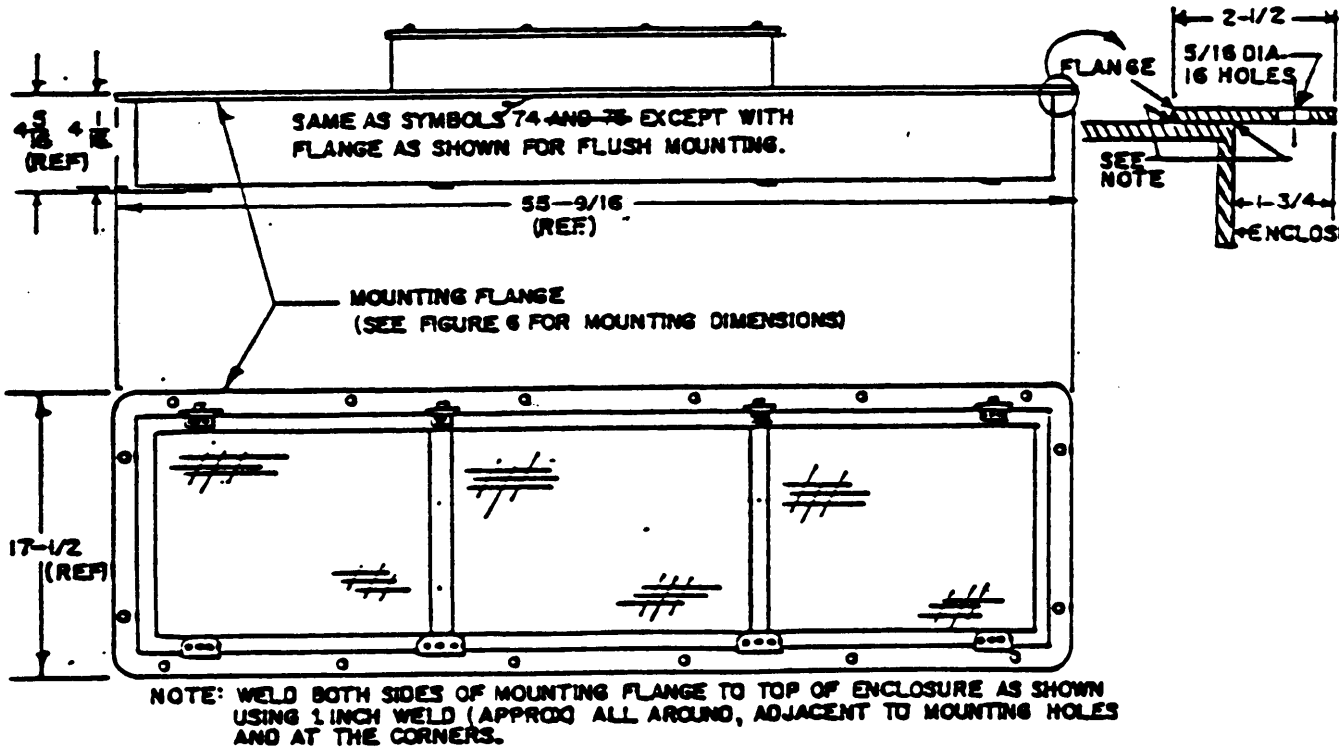


FIGURE 4. Typical views for Symbol 74.1.

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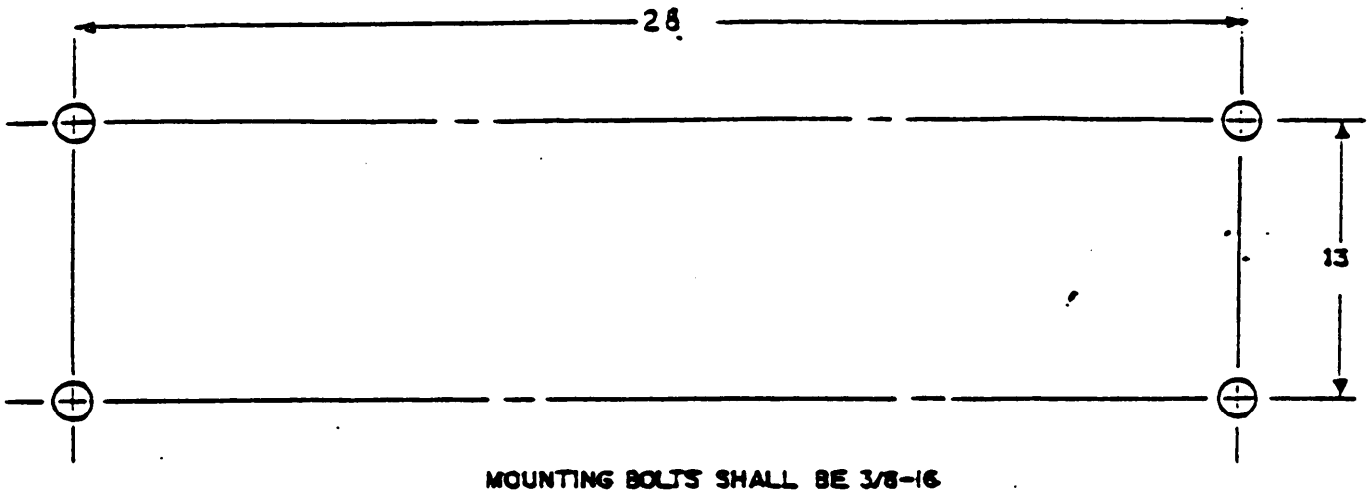


FIGURE 5. Mounting dimensions for Symbol 74.

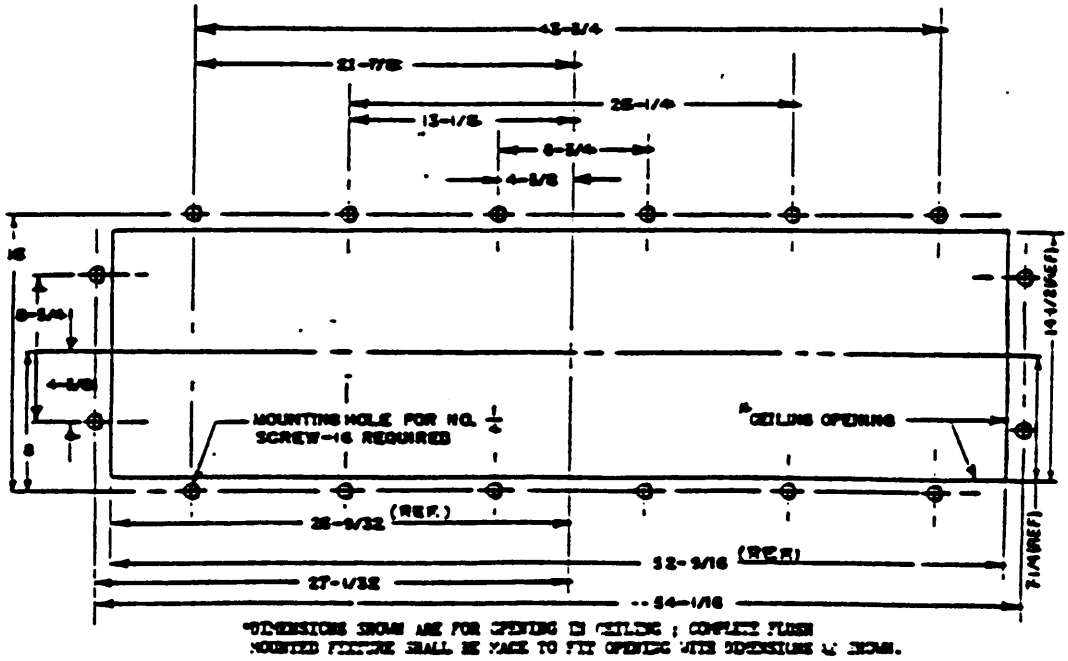


FIGURE 6. Mounting dimensions for Symbol 74.1.

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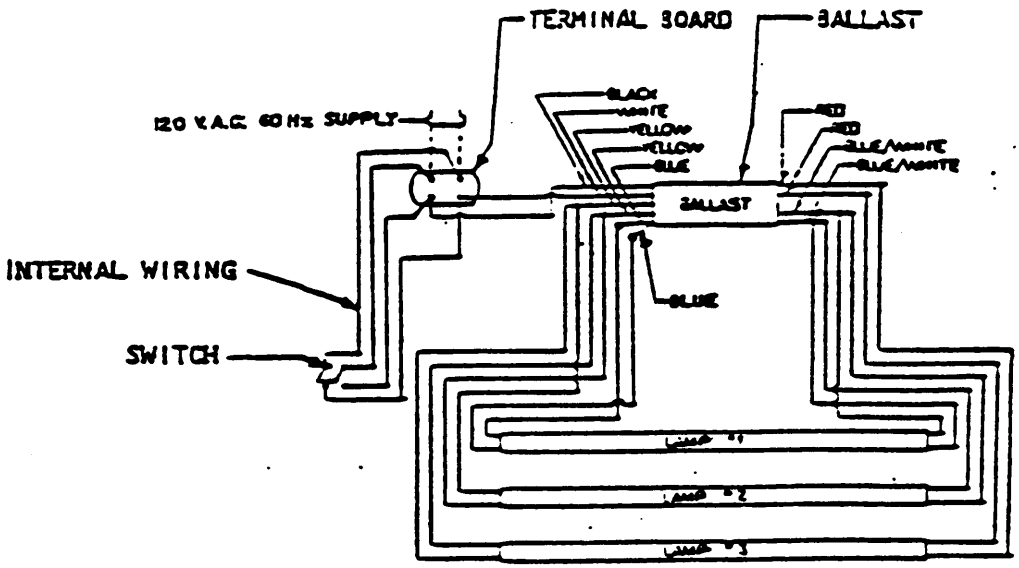


FIGURE 7. Wiring diagram.

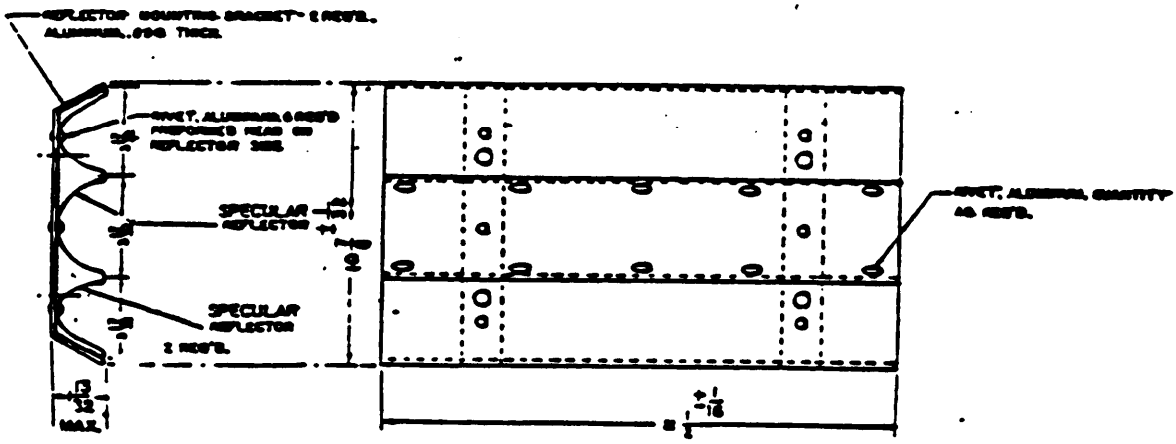
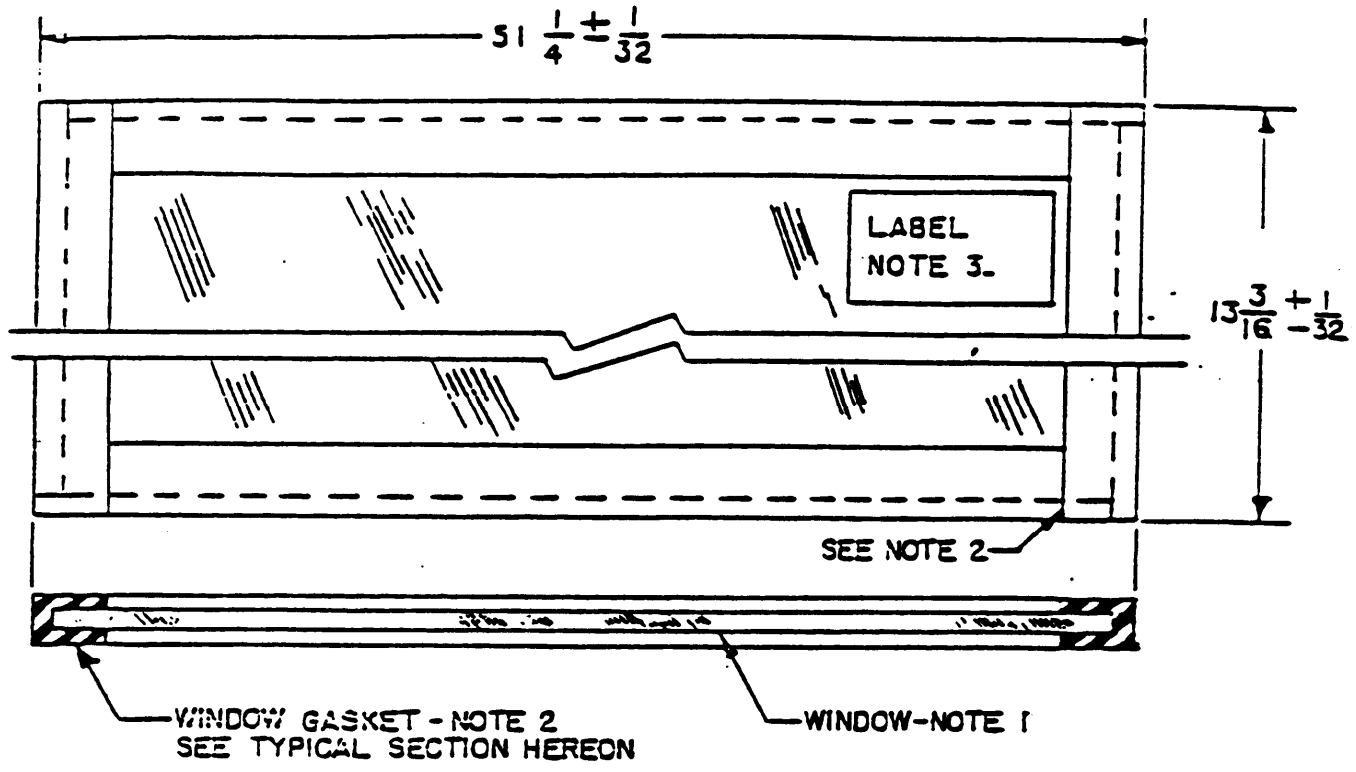


FIGURE 8. Reflector assembly.

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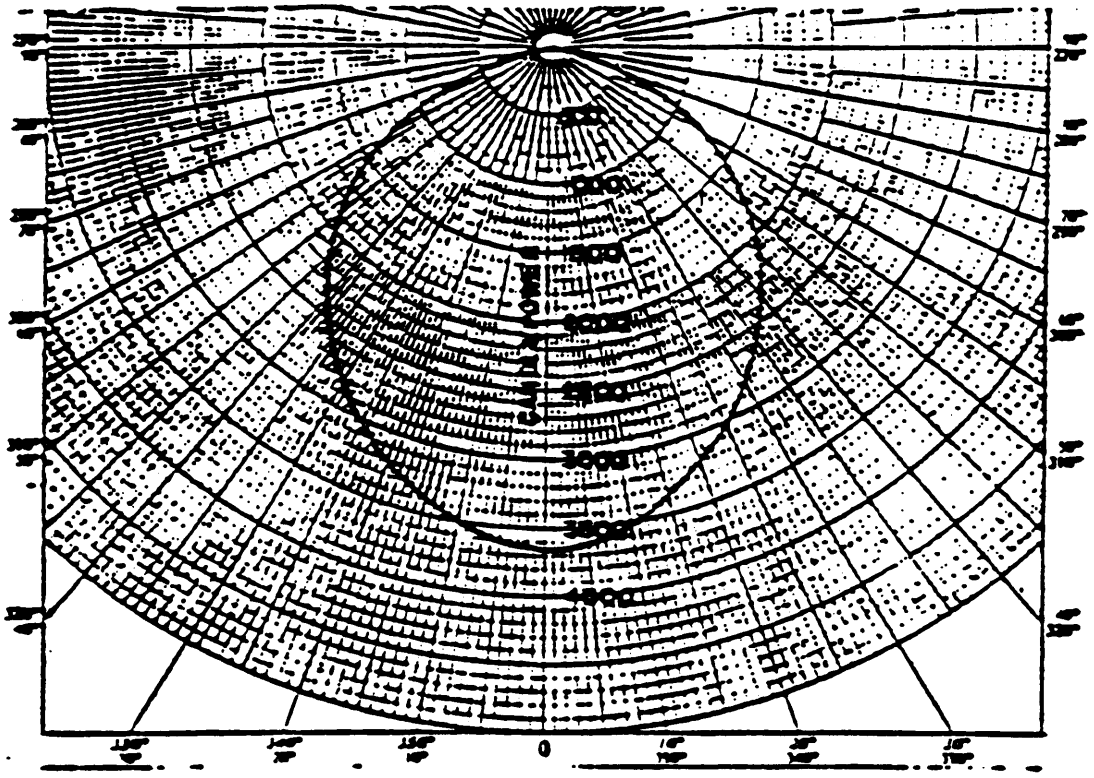


NOTES:

1. Material for window shall be acrylic plastic as follows:
   Part Number Ml 6377/ 13-001 for Symbols 74 and 74.1: Clear, prismatic (one side only), 1/8 inch thick, type CLEAR-2 of MIL-P-24191.
2. Material for window gasket shall be neoprene rubber 25-35 durometer. Gasket shall be cemented to window. Gasket maybe furnished in segments.
3. For window assemblies furnished as repair parts, a type IV label shall be furnished in accordance with MIL-DTL-16377. Location and the information to be shown on this label shall be as shown hereon.
4. Window assembly shall be cemented to window frame.

FIGURE 9. Window assembly.

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ANGLE	SYMBOLS 74, 74.1	ANGLE	SYMBOLS 74, 74.1
0	3650	65	900
5	3600	70	550
10	3500	75	250
15	3350	80	0
20	3175	85	0
25	3025	90	0
30	2800	95	0
35	2575	100	0
40	2375	105	0
45	2125	110	0
50	1850	115	0
55	1575	120	0
60	1250	125	0

Photometric test data:

Distance: 18.3.
Plane: At right angles to long axis of fixture through geometric center of lamp.
Voltage: 115 volts, 60 Hz.
Ambient temperature: 76 F.
Lamps: Use lamps furnished with fixture seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp correction: Distribution curves based on 4300 lumens per lamp.

FIGURE 10. Candlepower distribution curves for Symbols 74 and 74.1.

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REQUIREMENTS:

CLASSIFICATION:
Type: I
Class: 2

SYMBOL NUMBERS:
74: Lighting fixture, standard mounting; clear prismatic window; for use with three 60 watt cool white lamps.
74.1: Lighting fixture, flush mounting, clear prismatic window, for use with three 60 watt cool white lamps.

DESIGN:
Electrical rating: 180 watts, 120 volts, 60 Hertz.
Dimensions and configuration: See figures 1 through 4.
Mounting dimensions:
Symbol 74: See figure 5.
Symbol 74.1: See figure 6.
Enclosure: Dripproof (tested as specified herein).
Cable: Ten feet of type LSDPS-3 in accordance with MIL-C-24643/26.
Wiring diagram: See figure 7.
Internal wiring: Wire shall be of stranded soft annealed copper of a cross section to provide ample and safe current-carrying capacity and mechanical strength. Only high temperature wire (105° Centigrade (C.) minimum) shall be used.
Window and gasket assemblies: See figure 9.
Lampholder spacing: After final assembly of stationary and movable lampholder brackets in operating position, the distance between bracket faces shall be 46.844 $\pm$ 0.060.
Lampholder support brackets: Lampholder support brackets shall be provided. At one end of housing this bracket shall be permanently secured. At the other end the lampholder support bracket shall have limited movement, controlled by quick attachment fasteners to permit removal and installing new lamps. In no case shall the lamps fall from fixture when movable support bracket is in loosened position.
Reflector assembly: See figure 8. Material shall be aluminum lighting sheet, 0.020-.025 thick, prefinished, specular.
Light output:
Symbol 74 and 74.1: See figure 10.
Switch, safety: DPST push button. Upon unlatching window and frame assembly, HI-voltage to lampholders and 120 a.c. voltage to ballast shall be cut by release of pressure on actuating button of switch.
Weight:
Symbol 74: 60 lbs. maximum.
Symbol 74.1: 67 lbs. maximum.

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Designation and marking: The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figure 2.

(a) Two type I.
(b) One type IV. This label shall provide clear and concise step by step procedures for the removal and replacement of lamps in the fixture.
(c) One type IV. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in table I and window assemblies of table IV by name and stock number only.

Standard parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table II.
Window assemblies:	See table III.
Conformance:	
Comparison inspection:	Same as the first article inspection.
Inspection of product for delivery:	
Examination:	See MIL-DTL-16377.
Examination for preparation of delivery:	See MIL-DTL-16377.
Conformance tests:	See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table IV.

TABLE I. Standard parts.

Name	Specification or Military part number	National stock number (NSN)	Remarks
Lamp: Sym. 74 and 74.1	MS18226-1	6240-00-933-5075	F48T12/CW/HO/HT
Lampholder	UL-542	6250-00-906-8084	SUPP. SA, FIG. SA2.3
Ballast		6250-00-906-8083	1/
Shockmounts: External Internal	 M16377/41-026 M16377/41-028	6210-00-396-1717	

See footnote at top of next page.

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1/ Ballast shall be designed for starting and operating three 60 watt lamps, commercial designation F48T12/HO. Ballast shall conform to the following:

(a) Ballast shall be Underwriters Laboratories (UL) approved for class P (thermally protected) with automatic-resetting (thermostatic) type in accordance with UL Standard UL 935.
(b) Ballast rating shall be:
Input voltage 120, 60 Hertz.
Maximum Line current 1.70 Amperes.
Maximum Input watts 195.
Power factor 90% minimum.
Nominal Amps 0.800.
Voltage Output 460.
Type Rapid start.
Minimum start temperature - 20°F.
(c) Mounting dimensions: Four mounting holes, 1/4 inch diameter, on 2 by 13-3/4 inches centers.
(d) Weight: pounds minimum 14 pounds maximum.

TABLE II. First article and comparison inspection for complete fixtures.

Tests	Requirement	Test method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14	Dripproof 1/
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	33 Hz maximum
Enclosure effectiveness	3.5.11	4.8.14.2	Dripproof 1/
Dielectric	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

See footnote at top of next page.

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1/ The fixture shall be mounted in normal operating position and energized for a period of 2 hours prior to, and throughout, the water test which shall be conducted as follows: A continuous spray (similar to that of a sprinkling can) at a distance of 1 foot shall be directed to successively cover all surfaces, while fixture is slowly rotated on the long axis alternately exposing all parts of the fixture (side, bottom, side, window, and so forth). Minimum duration of test shall be 5 minutes.

TABLE III. First article and comparison inspection for window assemblies.

Tests	Requirement	Test method	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.2	See note 1 of table II
Light output	3.7.1	4.8.6.1	Candlepower

TABLE IV. Part numbers.

Military part number M16377/13-	Item name	National stock number (NSN)
74	Complete fixture, Symbol 74	6210-00-474-5653
74.1	Complete fixture, Symbol 74.1	
001	Window, clear prismatic	
003	Window, amber prismatic	

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy - SH
(Project 6210-N661-20)

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